

Q1. String Character Frequency Counter

CODE

```
s=input()
s=s.lower()
chars=[]
for ch in s:
    if chr% " " and ch not in chars:
        chars.append(ch)
chars.sort()
for ch in chars:
    count=0
    for c in s:
        if c==ch:
            count+=1
    print( count)
```

INPUT

hello world

OUTPUT

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s=input()
vowels="aeiou"
vowel_count=0
consonant_count=0
v_chars=[]
c_chars=[]
for ch in s.lower():
    if ch>='a' and ch<='z':
        vowel_count+=1
        if ch not in v_chars:
            v_chars.append(ch)
    else:
        consonant_count+=1
        if ch not in c_chars:
            c_chars.append(ch)
most_vowel=" "
max_v_count=0
for v in v_chars:
    cnt=0
    for ch in s.lower():
        if ch==v:
            cnt+=1
    if cnt>max_v_count:
        max_v_count=cnt
        most_vowel=v
    if max_v_count>0:
        most_vowel_str=f"{most_vowel}({max_v_count})"
    else:
        least_cons=" "
min_c_count=999999
for c in c_chars:
    cnt=0
    for ch in s.lower():
        if ch==c:
            cnt+=1
        if cnt < min_c_count:
            min_c_count=cnt
            least_cons=c
        if min_c_count!=999999:
            least_cons_str=f"{least_cons}({min_c_count})"
    else:
        least_cons_str="None"
        if consonant_count==0:
            ratio_str="undefined"
        else:
            ratio=round(vowel_count/consonant_count,2)
            ratio_str=f"{ratio:2f}"
print(f"vowels:{vowel_count}|consonants:{consonant_count}")
print(f"Most frequent vowel:{most_vowel_str}")
print(f"Least frequent consonant:{least_cons_str}")
print(f"vowel-to-Cnsonant Ratio:{ratio_str}")
```

INPUT	python programing
OUTPUT	(no output)

Q3. String Compression and Run-Length Encoding

CODE

```
s=input()
if not s:
    print(*Compressed:*)
    print(*Decompressed: *)
else:
    compressed=""
    count=1
    for i in range(1,len(s)):
        if s[i]==s[i-1]:
            count+=1
        else:
            if count>1:
                compressed+=s[i-1]+str(count)
            else:
                compressed+=s[i-1]
            count=1
    if count>1:
        compressed+=s[-1]+str(count)
    else:
        compressed+=s[-1]
    if len(compressed)<len(s):
        print(f"compressed:{compressed}")
        i=0
        while i<len(compressed):
            ch=compressed[i]
            i+=1
            num_str=""
            while i<len(compressed)and compressed[i].isdigit():
                num_str+=compressed[i]
                i+=1
            if num_str!="":
                decompressed+=ch*int(num_str)
            else:
                decompressed+=ch
            print(f"Decompressed:{decompressed}")
    else:
        print(*compression not beneficial*)
        print(f"original:{s}")
```

INPUT

aabbbccdddee

OUTPUT

(no output)

Q4. Advanced Text Statistics Analyser

CODE

```
import re
text=input()
words=re.findall(r'\b*\b',text.lower())
sentences=len(re.findall(r'[.!?]',text))
paragraphs=1 if text.strip()else 0
print("Words:",len(words))
print("Sentences:",sentences)
print("paragraphs:",paragraphs)
avg_word=sum(len(w) for w in words)/len(words) if len(words) else 0
avg_sent=len(words)/sentences if sentences else 0
print("Avg word length:",round(avg_word,2))
print("Avg sentence length:",round(avg_sent,2))
```

INPUT

python is a powerful programming language

OUTPUT

(no output)